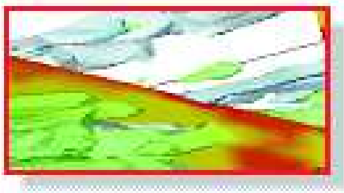
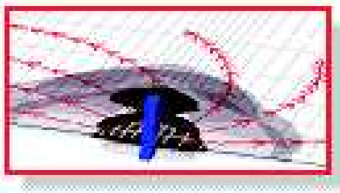
Modeling biogeochemical processes controlling groundwater quality in a small catchment

Step-by-step Instructions

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1 Overview

This exercise is a step-by-step modeling project, which we will progress over the first 4 days of the course at the end of each day. Each step corresponds and is focusing on the specific process that are being newly taught earlier on that day. Generally there will be detailed step-by-step instructions provided. As part of this project you will also learn how to compile a project-specific reaction database. These skills will prepare you for your own reactive transport modeling project in the future. The project involves a two-dimensional simulation problem that eventually includes

- advective-dispersive transport
- mineral reactions
- cation exchange and surface complexation, and
- kinetically controlled redox reactions.

The exercise is a conceptual modeling problem, meaning that we will not compare the simulation results to any measured data. The purpose of the exercise is mainly to illustrate how different processes individually and jointly affect groundwater quality evolution.

The problem is set up as a simple vertical transect representing a small catchment with groundwater recharge being the only driver for groundwater flow, i.e., there is no pumping or injection. The model extends all the way to the catchment boundary and all outflow occurs through groundwater discharge towards a river. We will study the evolution of the groundwater quality within the aquifer and how this affects the composition of the water that discharges to the river. The simulations will mimic, although in a crude way, how groundwater quality changes may occur in response to the increased atmospheric sulfur deposition (Xiao and Liu, 2002) that has occurred in the past, and in some places still occurs. Those elevated atmospheric sulfur deposition occurred in conjunction with air pollution from industrial emissions (Franken et al., 2009; Streets and Waldhoff, 2000). The increased sulfur input into the groundwater system is associated with a decline of the pH within the rainwater (i.e., acid rain). The increased acidity of the recharge water can cause the enhanced dissolution of selected minerals (Huang et al., 2019), which act as pH buffers. The dissolution of these minerals will raise the pH of the groundwater until they are depleted. With the exercise you will compare different scenarios and explore the influence of the initial concentration of the buffering mineral on the longevity of the buffering process. Moreover, the difference in composition between native groundwater and rainwater and the associated mineral dissolution will cause release and re-sequestration of ions that are originally associated with sediment cation exchange and/or surface complexation sites. As part of the the exercise you will incorporate the relevant cation exchange and surface complexation reactions and explore their influences on the groundwater quality evolution. Lastly, the rainwater also contains dissolved oxygen, which can trigger redox reactions if the receiving aquifer contains minerals that can act as reductants. For example, if pyrite is prevalent in a reducing aquifers, the intrusion of aerobic recharge water would oxidize the pyrite and potentially cause metal(loid) mobilization. As part of this exercise you will incorporate the kinetically controlled pyrite oxidative dissolution reaction and explore the influence of pyrite oxidation on groundwater quality.

2 Groundwater flow and conservative transport

2.1 Spatial discretisation and flow problem

The problem is set up as a simple two-dimensional flow and transport problem with dimensions and properties as defined in Table 1. To construct the MODFLOW model that underlies the transport problem proceed as follows:

- In ORTi, create a new model and save it in a new, separate directory. Remember to use only alphabets (either lower or UPPER case), numbers, underscore (_) and hyphen (—) in the path.
- Under Parameters → Model select the Model button and specify the attributes of the new model. Select the values Xsection for Dimension, unconfined for Type and MODFLOW series for Group
- Under Parameters → Model select the Grid button and specify the details of the model domain and the spatial discretization in the dialog that opens up. Define a vertical transect that is 2000m long and 20m thick. The model domain may be discretized into grid cells of 40m length and 1m height. Note, that the model input in ORTi is grid-independent and that therefore the model discretization can be easily changed at any time later.
- Select Parameters and select the Time button. Set the Total Simulation Time to 18250 days and select a time Step size of 365 days.
- Go to Spatial Attributes → MODFLOW → LPF → lpf.8 and set the Backg. value of horizontal hydraulic conductivity to 20 m/day. Then press OK to confirm the value.
- Go to Spatial Attributes → MODFLOW → LPF → lpf.9 and set the Backg. value of vertical hydraulic conductivity to 1 m/day. Then press OK to confirm the value.
- Go to Spatial Attributes → MT3DMS → BTN → btn.11 and set the Backg. value of effective porosity to be 0.25. Then press OK to confirm the value.
- Implement a hydraulic boundary condition that represents the river at the downstream end of this synthetic catchment by defining a specified head boundary at the downstream side of the model. First, define the type of boundary condition by going to Spatial Attributes → MODFLOW → BAS6 → bas.3 Boundary Conditions and setting the value to 1 (= active cell) for all cells. After that select Zone and use the Zone Tools to draw a short line near the effluent end. Choose an elevation of z = 8.5 m for the line and draw it from x = 1m to x = 20m.

The value for the line (=zone) needs to be set to -1. Use, for example, *river_bc* as Zone Names. You will see the selected name displayed near the location of the line.

By setting the boundary condition type to -1 the hydraulic head at the grid cells near the defined line will remain unchanged over the entire simulation period, i.e., it will remain at the value that corresponds to the elevation of the river water level. In the next step the initial heads need to be defined.

- First the define the global initial hydraulic head for the entire domain (Background value) by going to Spatial Attributes → MODFLOW → BAS6 → bas.6 Initial Heads and set the Backg. value to 20m. Then press OK to confirm the value.

- Define the initial hydraulic head at the location of the river by going to Spatial Attributes → MODFLOW → BAS6 → bas.6 Initial Heads. Select Zone and use the Zone Tools to draw again a short line near the effluent end. Choose an elevation of z = 8.5 m for the line and draw it from x = 1m to x = 20m. Note, that you can edit the coordinates manually after you have completed the drawing. The value for the line (=zone) needs to be set to 8.5. Use, for example, *river_head* as Zone Names. You will see the selected name and the selected value displayed near the location of the line.

  To define the groundwater recharge for this model you will first need to activate the recharge package, as it is not automatically activated. This can be done by going to Add-in → MODFLOW modules and ticking the box RCH. Then tick OK. You will see that you can now go to Spatial Attributes → MODFLOW → RCH → rch.2 Recharge value and set the Backg. value to 0.001m, corresponding to an annual groundwater recharge of 365mm per year. Then press OK to confirm the value.

2.2 Running MODFLOW

You have now completed to set up the flow model and you are ready to run MODFLOW.

- Under Parameters → 2.Flow use the Write Button to write all the input files required to run MODFLOW

- Under Parameters → 2.Flow use the Red Arrow button to start the MODFLOW simulation

- Check the simulated head contours and assess whether the results are plausible.

Table 1: Flow and transport parameters for the simulation.

Flow simulation	steady state
Total simulation period ($days$)	18250
Time step length ($days$)	182.5
Model length (m)	2000
Model thickness (m)	20
Grid spacing Δx (m)	40
Grid spacing Δz (m)	1
Porosity	0.25
Horizontal hydraulic conductivity (m/day)	20
Vertical hydraulic conductivity (m/day)	1
Piezometric head downstream boundary (River) (m)	8.5
Longitudinal dispersivity (m)	2.5
Transverse vertical dispersivity (m)	0.025

- Try, for example to set the head contours manually at 0.25m intervals, starting from 8.5m.

2.3 Conservative transport

For the simulation of the solute transport of the mobile components, the model parameters that control advective and dispersive transport need to be entered. You will proceed as follows:

- Go to Parameters → 3.Transport and select the Parameters button.
- In the dialog box that subsequently appears, select ADV → adv.1-Major data → MIXELM: Method → MMOC as the Solution Scheme. Leave the default parameters and click Apply and Close.
- Select Spatial Attributes → MT3DMS → DSP, enter Backg. longitudinal dispersivity value of 5 m and click OK.
- Select Spatial Attributes → MT3DMS → DSP, and enter Backg. value for dsp.3 ratio between horizontal transverse dispersivity and longitudinal dispersivity as 0.01. Click OK. This will define the actual value of horizontal transverse dispersivity to be 0.05 m (= 5 × 0.01 m).
- Add a tracer to the recharge water. This can be done by selecting Spatial Attributes → MT3DMS → btn.23 Recharge. Allocate an initial concentration of 10 mg/L to the recharge water. Click OK.

- Run the transport model by selecting Parameters → 3.Transport → Write. This will write all required MT3DMS input files. Then press the Red Arrow button to begin the transport simulation.
- Visualise the concentration contours under Results → 3.Transport → Tracer. You can change the Time Step (under Results) → 1.Model → Tstep to visualise the results for other simulation times. Also, after selecting a time it is possible to use the mouse wheel to changes times.
- Check the simulated tracer concentration contours and assess whether the results are plausible.
- Remember to save your ORTi file.
- End of Day 1.

Table 2: Aqueous concentrations used in acid groundwater recharge exercise

Aqueous component	C_{init} $(mol\ l_w^{-1})$	$C_{recharge}$ $(mol\ l_w^{-1})$
pH	7.00	4.18
pe	-3.00	14.00
Al	1.538×10^{-5}	0.0
C(4)	6.041×10^{-5}	5.736×10^{-5}
C(-4)	0.0	0.0
Ca	9.000×10^{-4}	2.430×10^{-4}
Cl	3.000×10^{-3}	7.000×10^{-4}
Fe(2)	1.900×10^{-5}	0.0
Fe(3)	0.0	0.0
K	5.000×10^{-5}	2.769×10^{-5}
Mg	1.600×10^{-4}	8.670×10^{-5}
Na	1.000×10^{-3}	2.827×10^{-5}
O(0)	0.0	5.000×10^{-4}
P	1.000×10^{-5}	0.0
S(6)	0.0	1.072×10^{-5}
S(-2)	1.000×10^{-5}	0.0
Si	1.107×10^{-4}	0.0
Zn	4.000×10^{-7}	0.0

3 Multi-species Conservative Transport and Mineral Dissolution

3.1 Data input for reactive transport

In the following step we will now activate the geochemical reaction network and define the various water and mineral compositions that will play a role in this simulation problem. The initial and the groundwater recharge water composition that will be employed in this example are shown in Table 2.

- Open your ORTi model from Day 1.
- Add a database file that is going to be used for the PHT3D simulation into the folder that contains all files for this exercise and name the file pht3d_datab.dat. Note that the database has to have this name in order for PHT3D to recognize it. You can use the standard PHREEQC database (phreeqc.dat) that will be provided for this exercise.
- Go to Parameters → 4.Chemistry and select the Import database button. ORTI will then read and interpret the PHT3D database file.

- The initial and the groundwater recharge water composition that will be employed in this example are shown in Table 2 below. Note, that you should always check the charge balance in PHREEQC before using the concentrations in the reactive transport simulations. In the present example you will indeed need to charge balance the solutions in PHREEQC and reduce the charge balance error to 0 %. Use chloride to charge balance your solutions. In addition we assume that the ambient groundwater was in equilibrium with two selected minerals at the time of the start of the simulation. To achieve this, equilibrate the background solution with $Quartz$ and with $Al(OH)_3$.

- To define initial concentrations of all aqueous components and also the concentrations of the acidic groundwater recharge, click on (Ad_Chemistry) and go to the Solutions Tab. The composition of the ambient water (initial water composition at the start of the simulation) needs to be entered in the Background column. Activate all relevant species by ticking the appropriate boxes for Al, C(4), C(-4), Ca, Cl, Fe(2), Fe(3), K, Mg, Na, O(0), P, S(-2), S(6), Si, Zn, and, as always, pH and pe. Enter the concentrations (after you have charge balanced the solutions). Similarly, define the acid recharge water composition in the Solution 1 column.

- The Background water composition will automatically be used to define the initial concentrations.

- The allocation of Solution 1 groundwater water recharge composition can be done by selecting Spatial Attributes → PHT3D → ph.5 Recharge and set Backg. to 1.

- Click OK.

- Remember to save your ORTi file.

3.2 Running PHT3D

In this first step we will now run PHT3D without invoking any mineral reactions. This means that all temporal and spatial concentration changes of the aqueous components that occur over the simulation period are solely a function of advective-dispersive transport. This provides the basis for subsequently studying the impact of mineral reactions.

- Go to Parameters → 4.Chemistry and use the Write Button to write all the input files required to run PHT3D

- Then go to Parameters → 4.Chemistry and use the Red Arrow button to start PHT3D.

3.3 Visualization of results

After the end of the simulation visualize the results in various ways.

- Obtain a contour plot by navigating in the Results panel of ORTI. Select a specific timestep and a species for which you want to see your results. For example to see the pH contours after 10 years select Results → 1.Model → Tstep and select 3650 and then go to Results → 4.Chemistry → Species and select pH. You can easily move forward and backward in time by clicking Tstep once more and then using your keyboard arrows to move forward or backward.

- Plot breakthrough curves at the river discharge point to illustrate how the quality of the water discharging from the groundwater to the river changes over time. First, define the location for which you want to plot breakthrough curves by going to Spatial Attributes → Observation → OBS → obs.1 → Add Zones → Add Point(+) and use your mouse to position the cursor and clicking at the location where you wan to create an observation point. In the dialog that comes up name the zone $BTC1$. No specific input is required for the Zone Value field. The coordinates of the location selected by the mouse click can be modified, if required.

- Then, in the panel Results → 5.Observation, select Time-series Graphs and Chemistry. In the dialog select By Zone, tick on $BTC1$ in the Zones Tab, then tick on any layer (Layers Tab), then the chemical species of interest (Species Tab), and finally, click twice on Apply.

- Alternatively you can visualize results with PHT3DPlot.

- What changes occur to the chloride concentrations and what does this mean?

- How much does pH decrease over the duration of the simulation?

- Save the simulated breakthrough curves for pH, Si and Al so you can later directly plot and compare these results together with the results from your next model variants.

3.4 Role of mineral reactions

- Let's first study the effect of including *Quartz* in this simulation. This can be done under Parameters → 4.Chemistry → Ad_Chemistry → Phase Tab. The input allows to activate mineral phases, define their initial concentration and to define a value for the saturation index (SI) to which the mineral should be equilibrated (typically 0). For *Quartz* leave the Backgr_SI at the default value of 0 and define a background concentration Backgr_Mol of 1 mol/L_bulk.

- Now rerun PHT3D and inspect again the breakthrough curves for pH, Si and Al.
- How can you interpret these results?

Now we will inspect the impact of $Al(OH)_3$ (an amorphous aluminum hydroxide) on the composition of the discharging groundwater.

- Add initially a relatively small initial concentration such as 1×10^{-5} mol/L_bulk of $Al(OH)_3$, rerun the model and save again the breakthrough concentrations that you obtained for the observation point at the river.
- Increase the initial concentration to a value such as 1×10^{-2} mol/L_bulk of $Al(OH)_3$, rerun the model and save again the breakthrough concentrations that you obtained for the observation point at the river.
- Make comparative plots for pH and Al for your different model variants.
- How can you interpret these results?
- Remember to save your ORTi file.
- End of Day 2.

4 Cation Exchange and Surface Complexation

Due to the difference in composition, especially pH, between native groundwater and rainwater, and the associated dissolution of $Al(OH)_3$, which releases Al^{3+} and destructs sediment surface sites, the passage of the recharge water in the aquifer will lead to a series of exchange and adsorption/desorption processes. In the following step we will now activate the cation exchange and surface complexation reactions and explore their role in this simulation problem.

4.1 Cation Exchange

To include the cation exchange process in the reaction network, you will proceed as follows:

- Open your ORTi model from yesterday, plot and save the breakthrough curves for the concentrations of some of the cations (e.g., Al, Ca, Mg, K and Na).
- To add the exchanger species to the reaction network, go to Parameters → 4.Chemistry → (Ad_Chemistry) and activate X- under the Exchange Tab. Add a background value of 0.01 for the cation exchange capacity (CEC).
- Now rerun PHT3D.
- Once the model run is complete, plot again the breakthrough curves for the Al, Ca, Mg, K and Na concentrations, and compare the current results with previous modeling results without cation exchange.
- You can also study the impact of CEC on the reactive transport behaviour by rerunning the model with a different CEC.
- Remember to save your ORTi file.

4.2 Surface complexation

The default PHREEQC database (i.e., the database you have been using in this exercise so far) uses surface data from Dzombak and Morel (1990), which were developed for hydrous ferric oxide. In this simulation problem, the sediment surface sites are assumed to be provided dominantly by Al(OH)3(a). Therefore, to include the adsorption/desorption processes in the reaction network, you will first need to add the surface complexation reactions between aquoeus species and the $Al(OH)_3$ surface into the database.

While there is not a comprehensive literature database on surface complexation reactions for $Al(OH)_3$, data for surface complexation reactions with gibbsite (a crystalline aluminum hydroxide) are available. These are described in detail in Karamalidis and Dzombak (2010).

We will use gibbsite surface data as approximation in this exercise. Copy the relevant tables with surface reaction data from Karamalidis and Dzombak (2010) for this exercise from the wechat group. At this point, you should know how and where to add those reactions into the database.

- Open your database file
- Define one new surface master species of Gb/GbOH and have a try. You only need to include the reactions that are relevant, but you can of course include all the reactions in those tables if there is time. (If you accidentally make any error you can always make a new copy of the file phreeqc.dat and restart.) We will then send out our database such that you can compare and see if your reactions are correctly defined.

Once the surface complexation reactions are ready in the database, you will proceed as follows:

- Go to Parameters → 4.Chemistry and select the Import database button. ORTi will then read and interpret the new PHT3D database file.
- Go to Parameters → 4.Chemistry → Ad_Chemistry and go to the Surface Tab.
- Now, activate the surface with the name Gb. Enter Al(OH)3(a) and equilibrium under name and switch, respectively, to link the surface site to the concentration of Al(OH)3(a) in the aquifer, which is an equilibrium mineral. Enter 1.0×10^{-3} as the surface site density (unit: mole of sites per mole of mineral) under Site_back. After we define this concentration of surface sites as background concentration, it will be applied throughout the model domain. The initial occupancy of the surface sites will be automatically determined at the start of the simulation, i.e., there will be an equilibration step that makes sure that the aqueous composition(s) and the surface(s) are in equilibrium. The presence of the surface sites will cause the aqueous species to undergo surface complexation reactions, with the variable extent of the sorption being a function of the spatially and temporally variable geochemical conditions.
- The present model is using the no_edl option, which means that the surface complexation reactions do not consider any electric double layer effects. When you use the no_edl mode, no input is required for surface area and sorbent mass. If required in the future for your

own research project, the no_edl option can be changed under Parameters → 4.Chemistry → Parameters → PH → ph.6- specific options where a different option such as -ddl could be entered.

- Now rerun PHT3D and investigate how much the adsorption/desorption reactions have impacted the transport behaviour of the relevant species. Not all the species you added to the surface reactions will show differences because of the geochemical condition studied here, but some will.
- You can also study the impact of the surface site density on the reactive transport behaviour by rerunning the model with a different site density. The number 1.0×10^{-3} you used is actually very low. Now add a 100-fold increased surface site density of 0.1, rerun PHT3D and inspect the simulation results.
- Remember to save your ORTi file.
- End of Day 3.

5 Kinetically Controlled Redox Reactions

5.1 Pyrite oxidation

Due to the presence of dissolved oxygen in the rainwater, redox reactions such as pyrite oxidation will occur in the reducing aquifer. Pyrite oxidation will be defined as a kinetic process in which the reaction rate depends on the amount of dissolved oxygen in the recharge water. In a first step, we will include the rate expression of pyrite oxidative dissolution in the database, under the RATES block. The default phreeqc.dat has already had a rate expression for pyrite oxidation. Here, we will update the rate expression to the one below, to illustrate how to change/compile a rate expression in the database. This rate expression was originally proposed by Williamson and Rimstidt (1994), further updated by Appelo et al. (1998) and Prommer and Stuyfzand (2005), and used in many more recent studies, including that of Seibert et al. (2016), Rathi et al. (2018) and Prommer et al. (2018).

```
#########################
Pyrite
#########################
-start
10 if (m <= 1e-12) then goto 200
20 if (si("Pyrite") >= 0) then goto 200
30 rate_oxy = (mol("O2"))^ parm(1)
40 surf_over_vol = log10(m * parm(2))
50 f1=(mol("H+"))^parm(3) * 10^(-10.19+surf_over_vol)*(m/m0)^parm(4)
60 moles = rate_oxy * f1 * time
70 if (moles > m) then moles = m
100 put(moles,13)
200 save moles
-end
```

To include pyrite oxidation reaction in the model, you will proceed as follows:

- Open your ORTi model from yesterday.
- Now we add pyrite into our reaction network. Before you do this, check again your background solution in PHREEQC and if it is in equilibrium with pyrite. If it is over- or undersaturated equilibrate the solution and check what has changed. Update the water composition that defines the background water composition. To activate pyrite go to the Parameters → 4.Chemistry → Ad_Chemistry → Phases Tab. From the various listed minerals, select and activate pyrite. Use an initial concentration of 5.0×10^{-3} mol/L_bulk volume.

- Define the reaction rate parameters for pyrite oxidation under Parameters → 4.Chemistry → Ad_Chemistry → Kinetic_Minerals Tab. First, activate the pyrite kinetics by ticking the first (C) box. Add values of 0.5, 115, -0.11, and 0.67 for parm(1), parm(2), parm(3) and parm(4), respectively, corresponding to the 4 parameters that have been defined in the rate expression in the database.

- Rerun PHT3D and investigate how much the pyrite oxidation reaction has impacted the reactive transport behaviour of species of interest (e.g., Fe related species).

- Because the ferric Fe produced from pyrite oxidation is commonly assumed to rapidly precipitate as ferrihydrite (simplified as Fe(OH)3(a)), now also activate Fe(OH)3(a) (equilibrium version) under Parameters → 4.Chemistry → Ad_Chemistry → Phases Tab. Leave the initial concentration of Fe(OH)3(a) to be zero.

- Rerun PHT3D and investigate how much the ferrihydrite precipitation has impacted the reactive transport behaviour of species of interest (e.g., Fe related species).

- Note that the oxidation rate of pyrite is, among other factors, depending on the pyrite concentration. Where pyrite is present at higher concentrations the reaction will proceed faster. You can study the impact of pyrite initial concentration on the reactive transport behaviour by rerunning the model with a different pyrite initial concentration.

- Please also note that due to the neoformed ferrihydrite which has very high surface area, the sediment surface sites are probably no longer solely provided by Al(OH)3(a). You may have noticed that the more processes you add, the longer the model run takes. You can add the ferrihydrite sites to see the effects in your own time if you want. For the sake of time, we do not include ferrihydrite sites here to simplify the exercise for illustration purpose.

- Remember to save your ORTi file.

5.2 Heavy metal release associated with pyrite oxidation

Trace metals including zinc, cadmium and arsenic are often incorporated in the structure of pyrite and have been shown to co-dissolve stoichiometrically during pyrite oxidation. Here, we will model zinc co-dissolution to illustrate how the process works. In a first step, we will include the rate expression of dissolution of sphalerite (ZnS) in the database, again under the RATES block. The rate expression that is used for sphalerite will simply be linked to the rate of the computed pyrite oxidation. The stoichiometric ratio at which sphalerite will be dissolved, compared to pyrite, is determined by a constant. For example, if the constant is set to 0.001, the dissolution of 1 mmol pyrite will be accompanied by the dissolution of 1 μmol sphalerite. Please define the rate expression for sphalerite dissolution yourself. We will then send out our database such that you can compare and see if your rate expression is correctly defined. We do now expand the last simulation and include additionally the kinetic mineral sphalerite in the reaction network.

- Go to Parameters → 4.Chemistry and click on Import database button. ORTi will then read and interpret the new PHT3D database file.
- Activate the mineral sphalerite under Parameters → 4.Chemistry → Ad_Chemistry → Phases Tab and set the initial concentration for sphalerite to 5.0×10^{-3} mol/L_bulk.
- Define the ratio of sphalerite / pyrite dissolution to be 0.1 (i.e., set parm(1) to 0.1 for sphalerite) under the Parameters → 4.Chemistry → Ad_Chemistry → Kinetic_Minerals Tab. This number conceptually represents that within pyrite there is a 10% substitution by Zn.
- Now rerun PHT3D and investigate how much the sphalerite co-dissolution reaction has impacted the reactive transport behaviour of species of interest (e.g., Zn).
- Remember to save your ORTi file.

Finally, if there is still time, please define the co-dissolution of another metal sulfide mineral in the model and explore the metal release yourself. For example, you can try arsenic or cadmium. Please note that the default phreeqc.dat may not contain all the reactions you need depending on which element you decide to explore. However, you may find the relevant reactions in other PHREEQC databases (e.g., wateq4f.dat and minteq.v4.dat) and/or in the literature.

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